

# Born too soon: lifelong kidney risks and the importance of early intervention

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Premature birth and low birth weight disrupt kidney development substantially, which leads to an increased risk of acute kidney injury, chronic kidney disease and hypertension. Broader awareness of these risks, implementation of preventive strategies and optimization of neonatal care are key to safeguarding kidney health and ensuring better long-term outcomes.

Approximately one in ten neonates is born prematurely (before reaching 37 weeks of gestation (WG)), with notable variability worldwide. Prematurity accounts for 18% of under-five child mortality and 35% of neonatal deaths, which makes it the leading cause of mortality in these age groups<sup>1</sup>. Most these deaths occur in low- and middle-income regions, and many are preventable. In high-resource settings, survival of extremely preterm infants born as early as 23–24 WG continues to increase, and survival might even be possible for infants born at 22 WG. Advances in neonatal care, including the use of antenatal steroids, surfactant therapy, non-invasive respiratory support, parenteral nutrition and individualized care have improved survival<sup>2</sup>. However, preterm infants remain at a high risk of neurodevelopmental impairment, cognitive deficit, chronic lung disease and growth failure.

Preterm birth affects the development of nearly every organ system (particularly the brain, lungs, heart, eyes, gastrointestinal tract and the kidneys), which has long-term health implications. Although neonatal care primarily addresses the respiratory, cardiovascular and neurodevelopmental needs of preterm infants, both in the neonatal period and throughout long-term follow-up, the kidney consequences of prematurity are often underappreciated<sup>3</sup>.

Glomerulogenesis – the process of glomerular capillary tuft formation and the differentiation of podocytes, mesangial cells and glomerular capillary endothelial cells – is completed by 34–36 WG in humans. This process accelerates at ~20 WG and nearly 60% of nephrons are formed during the third trimester<sup>4</sup>. Consequently, very preterm infants might have a limited nephron reserve, which can be further reduced when prematurity is associated with intrauterine growth restriction. In children born prematurely, nephrogenesis does not entirely cease at birth; rather, it continues during the postnatal period, albeit with the formation of nephrons with morphological abnormalities, which reflects a deviation from the typical structural development seen during prenatal nephrogenesis. These nephrons, compared with those of term neonates at the same postmenstrual age (that is, gestational age plus chronological age), show enlarged glomerular diameter and tubular cell hypertrophy, which reflects early nephron stress and results in heightened vulnerability to kidney insults<sup>5</sup>. In addition, infants who are born preterm or with low body weight (LBW) – now referred to as small vulnerable neonates – are

frequently admitted to neonatal intensive care units (NICUs) where they are exposed to a variety of factors that can cause kidney injury, including episodes of acidosis, hypoxia and hypotension. In addition, these infants are often treated with nonsteroidal anti-inflammatory drugs and/or aminoglycosides, both of which can further compromise their immature kidneys. Nephron endowment in preterm neonates is thus particularly challenged by impaired nephrogenesis and the additional loss of vulnerable nephrons under NICU conditions.

Reduced nephron endowment in preterm infants has substantial clinical implications. Premature infants are at elevated risk of acute kidney injury (AKI) – in addition to impaired nephrogenesis, reduced renal blood flow compounds the likelihood of AKI. Moreover, the physiological immaturity of the kidney tubular system in neonates results in limited urinary concentration capacity, reduced sodium reabsorption and loss of solutes. Collectively, these mechanisms explain the substantial risk of pre-renal AKI in neonates, particularly in those admitted to NICU or requiring cardiac surgery. In patients born between 22 and 29 WG, the incidence of AKI approaches 50%<sup>6</sup>.

**“small vulnerable neonates require long-term, comprehensive follow-up focused on preserving kidney function”**

Oligonephronia, which is characterized by a reduced number of nephrons at birth, also triggers compensatory mechanisms that include glomerular hypertrophy and hyperfiltration. These adaptive changes might initially sustain kidney function but often lead to maladaptive consequences, such as focal segmental glomerulosclerosis and progressive proteinuria, that further diminish the already limited nephron reserve. This reduction in nephron endowment is strongly linked to an elevated risk of hypertension in adulthood<sup>7</sup>, which is probably due to increased intraglomerular pressure and sustained hyperfiltration stress. Small vulnerable neonates are particularly susceptible to these pathophysiological processes, which exacerbate their risk of developing cardiovascular–kidney–metabolic syndrome, a constellation of conditions that further compromises kidney health. Consequently, these early-life factors substantially increase the likelihood of chronic kidney disease (CKD) later in life for individuals born with LBW or prematurely.

Extensive epidemiological evidence links preterm birth and LBW with CKD risk. For example, a longitudinal study in Sweden that followed over 4 million live births reported an incidence of CKD of 0.1%, and preterm birth emerged as an independent risk factor for CKD that persisted from childhood into adulthood<sup>8</sup>. Similarly, data from Norway indicate that both LBW and being small for gestational age are associated with an increased risk of progression to kidney failure within the first 40 years of

life<sup>9</sup>. These findings underscore the long-term kidney vulnerability conferred by early developmental challenges and highlight the crucial need for early monitoring and preventive strategies in populations at risk.

Given these heightened risks, small vulnerable neonates require long-term, comprehensive follow-up focused on preserving kidney function. Assessing glomerular filtration rate in neonates and small infants presents unique challenges, which can hinder early detection of kidney injury. However, without early intervention, these children might miss opportunities to slow kidney disease progression. For many infants born prematurely or with LBW, clinical manifestations of oligonephropathy only become evident in childhood. Early identification largely depends on clinical judgment and a careful review of their neonatal history. Primary care providers should consider factors such as maternal hypertensive disorders, gestational age, birth weight and any episode of neonatal AKI to identify infants who might require closer follow-up. Neonatal intensive care practices have a pivotal role in shaping the long-term kidney health of at-risk neonates. Increased awareness among neonatologists of strategies to minimize kidney toxicity, as well as education on long-term renal risks, is vital to ensuring appropriate follow-up. This vigilance, however, must be balanced to prevent overdiagnosis and unnecessary medicalization.

In 2017, the LBW and Nephron Number Working Group proposed a lifelong kidney monitoring protocol for small vulnerable neonates<sup>10</sup>. The recommended strategy involves regular monitoring of blood pressure and urinalysis, with frequency adjusted on the basis of age, degree of prematurity and other neonatal risk factors, such as a history of AKI. Following a protocol similar to that used for solitary kidney monitoring, assessments should begin at three years of age (or even earlier for very preterm infants or those with neonatal AKI), continue every two years until age 40, and annually thereafter. If elevated blood pressure, proteinuria or other risk factors are detected, further kidney function evaluation using creatinine and cystatin C measurements should be conducted. Educating families about lifestyle modifications and avoiding nephrotoxic agents is a crucial aspect of this preventive strategy.

## “preterm birth and LBW carry substantial lifelong implications for kidney health”

A multidisciplinary approach to pregnancy counselling and specialized care is also recommended for preterm infants or infants with intrauterine growth restriction who have a family history of hypertensive pregnancy disorders. Adolescents with CKD who were born preterm might also benefit from this integrated care model, which aims to optimize maternal and offspring health, and reduces the transmission

of kidney risk factors across generations. In 2024, a Neonatal Kidney Health Consensus Workshop formulated recommendations for risk stratification and kidney health monitoring for NICU graduates<sup>11</sup>. The workshop identified three groups at high risk of long-term kidney health deterioration: preterm infants, critically ill infants with AKI and those with severe cardiac conditions. Based on these categories, the panel established consensus recommendations and outlined areas for future research.

In conclusion, preterm birth and LBW carry substantial lifelong implications for kidney health, which highlights the need for vigilant, long-term monitoring. Early detection, preventive strategies and informed neonatal care protocols are crucial to mitigate risks and improve health outcomes for this vulnerable population.

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Published online: 22 January 2025

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### Competing interests

The authors declare no competing interests.